Programming Fundamentals

Lab 3 & Lab 4

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Lab 3

Task 1:

Assume that the following variables are defined:

```
int age;
double pay;
char section;
```

Write a single cin statement that will read input into each of these variables.

Input

Task1.cpp

```cpp
#include <iostream>
using namespace std;

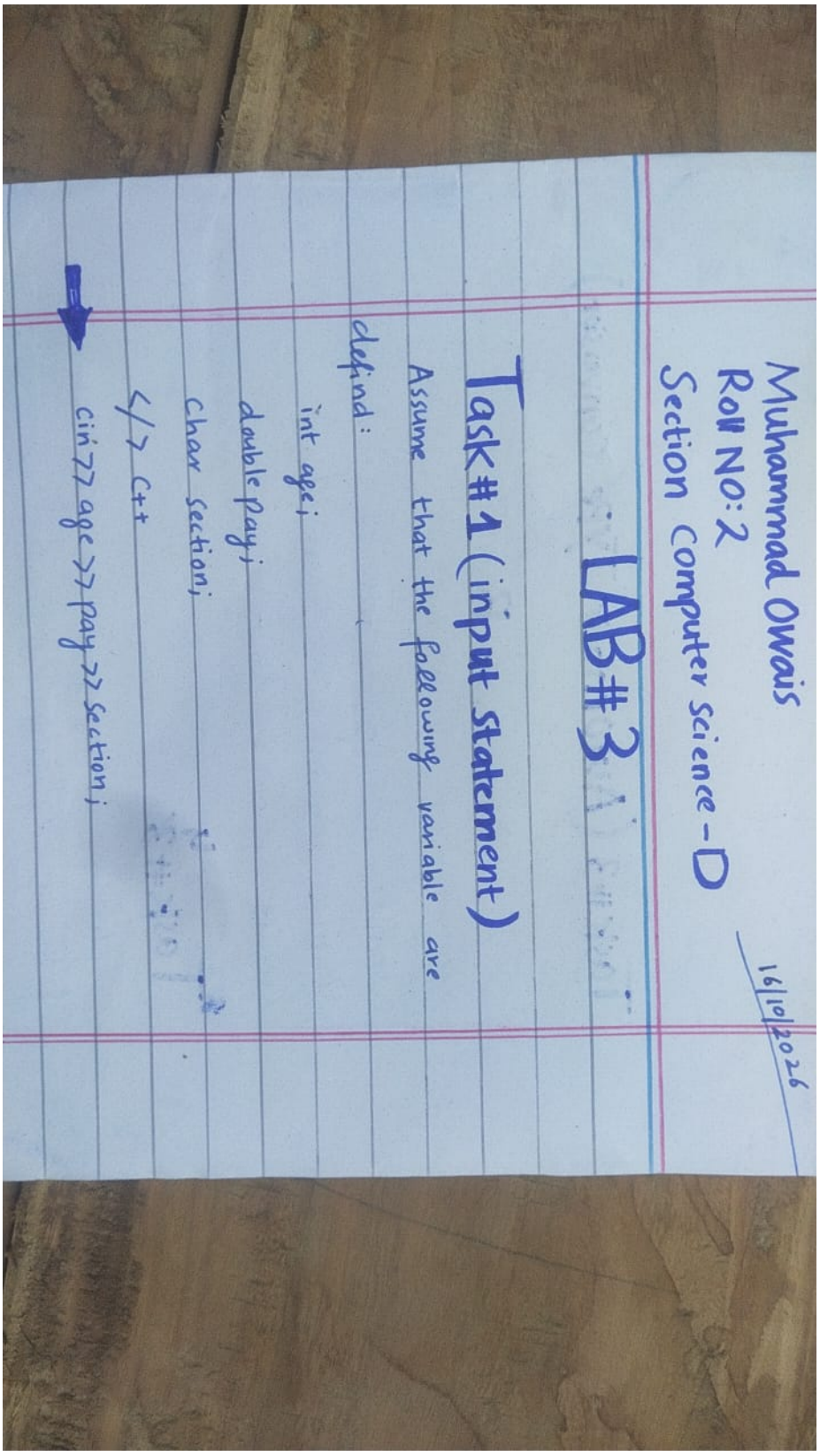
int main()
{
    int age;
    double pay;
    char section;

    cout << "Enter age, pay and section: ";
    cin >> age >> pay >> section;

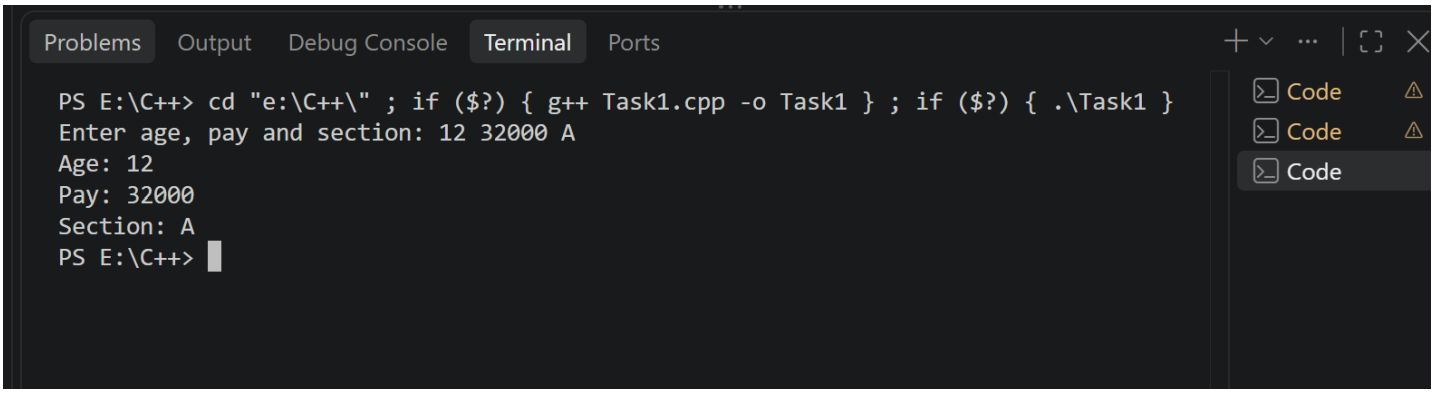
    cout << "Age: " << age << endl;
    cout << "Pay: " << pay << endl;
    cout << "Section: " << section << endl;

    return 0;
}
```

Handwritten Notes



Output


Problems
Output
Debug Console
Terminal
Ports
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ Task1.cpp -o Task1 } ; if ($?) { .\Task1 }
Enter age, pay and section: 12 32000 A
Age: 12
Pay: 32000
Section: A
PS E:\C++>
Code
Code
Code

Task 2:

Complete the following table by writing the value of each expression in the Value column according to C++ language rules.

Expression	Value
28 / 4 - 2	
6 + 12 * 2 - 8	
4 + 8 * 2	
6 + 17 % 3 - 2	
2 + 22 * (9 - 7)	
(8 + 7) * 2	
(16 + 7) % 2 - 1	
12 / (10 - 6)	
(19 - 3) * (2 + 2) / 4	

Input

Task1.cpp | Task2.cpp

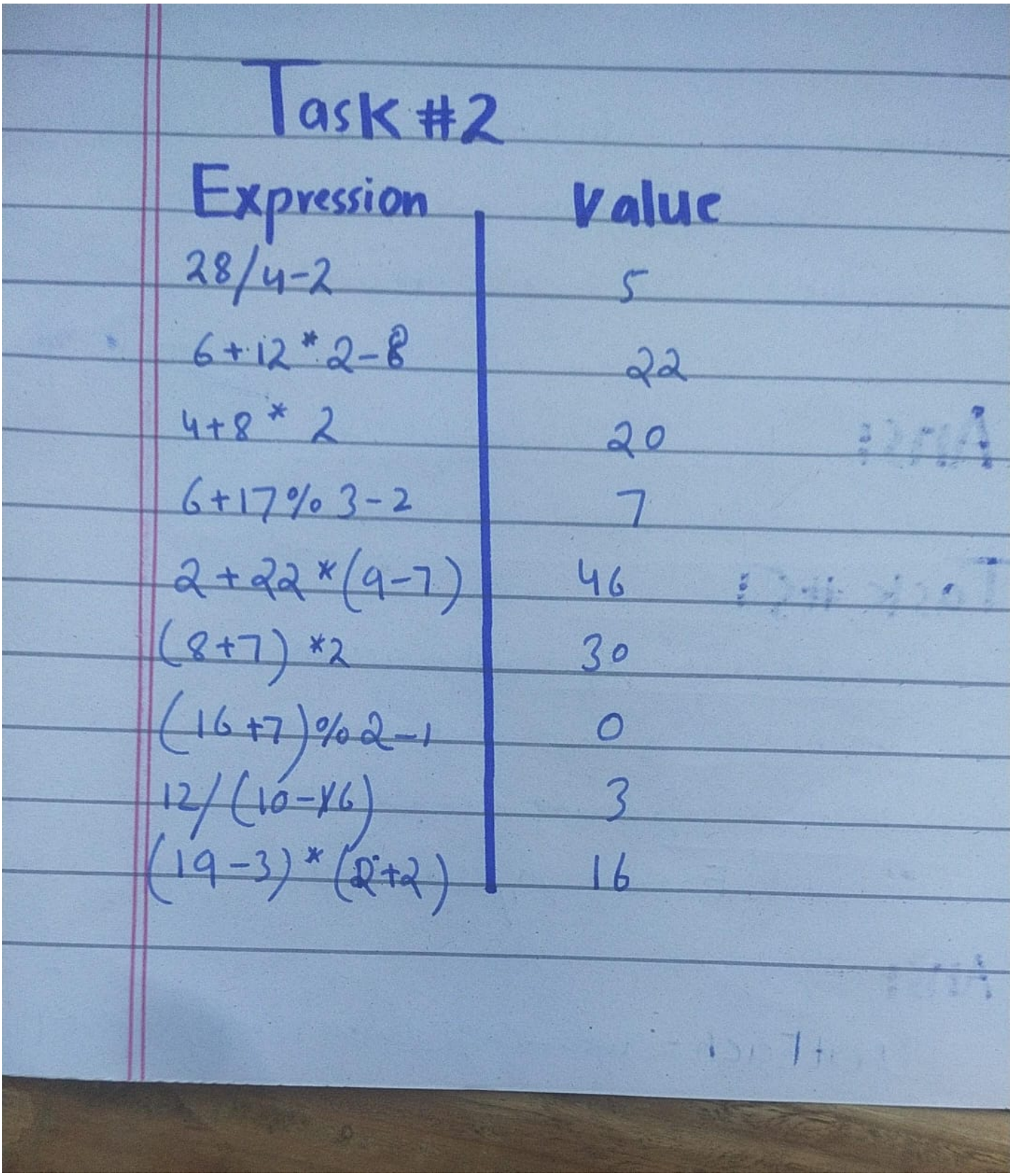
Task2.cpp > main()

```cpp
#include <iostream>
using namespace std;

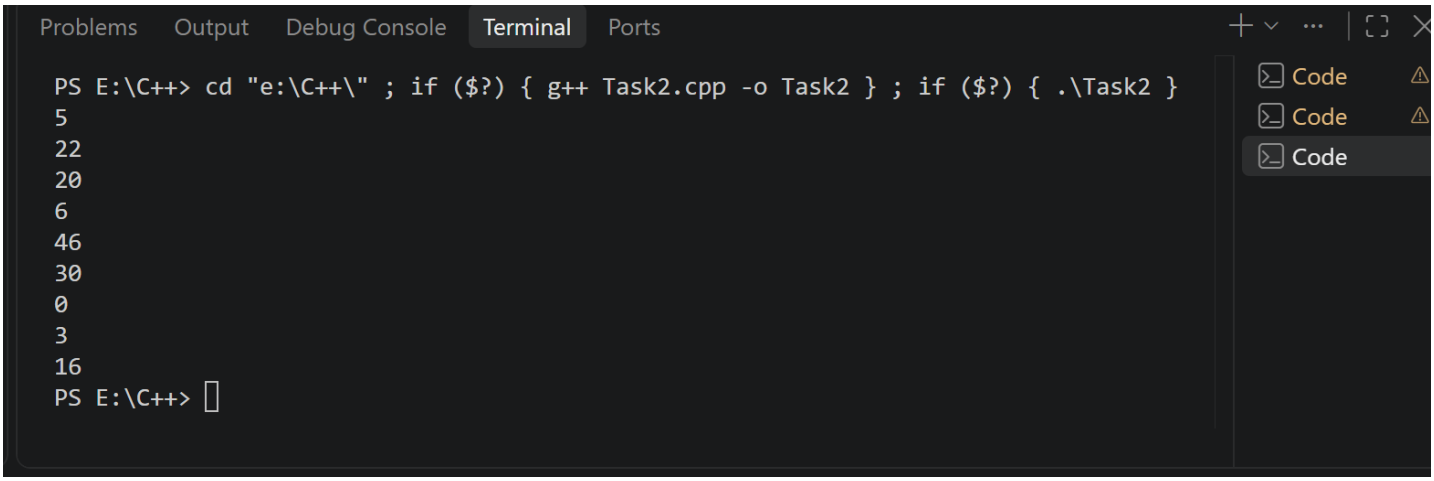
int main()
{
    cout << 28 / 4 - 2 << endl;
    cout << 6 + 12 * 2 - 8 << endl;
    cout << 4 + 8 * 2 << endl;
    cout << 6 + 17 % 3 - 2 << endl;
    cout << 2 + 22 * (9 - 7) << endl;
    cout << (8 + 7) * 2 << endl;
    cout << (16 + 7) % 2 - 1 << endl;
    cout << 12 / (10 - 6) << endl;
    cout << (19 - 3) * (2 + 2) / 4 << endl;

    return 0;
}
```

Handwritten Notes



Output


Problems
Output
Debug Console
Terminal
Ports
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ Task2.cpp -o Task2 } ; if ($?) { .\Task2 }
5
22
20
6
46
30
0
3
16
PS E:\C++>
Code
Code
Code

Task 3:

Assume a program has the following variable definitions:

```
int units;
float mass;
double weight;
weight = mass * units;
```

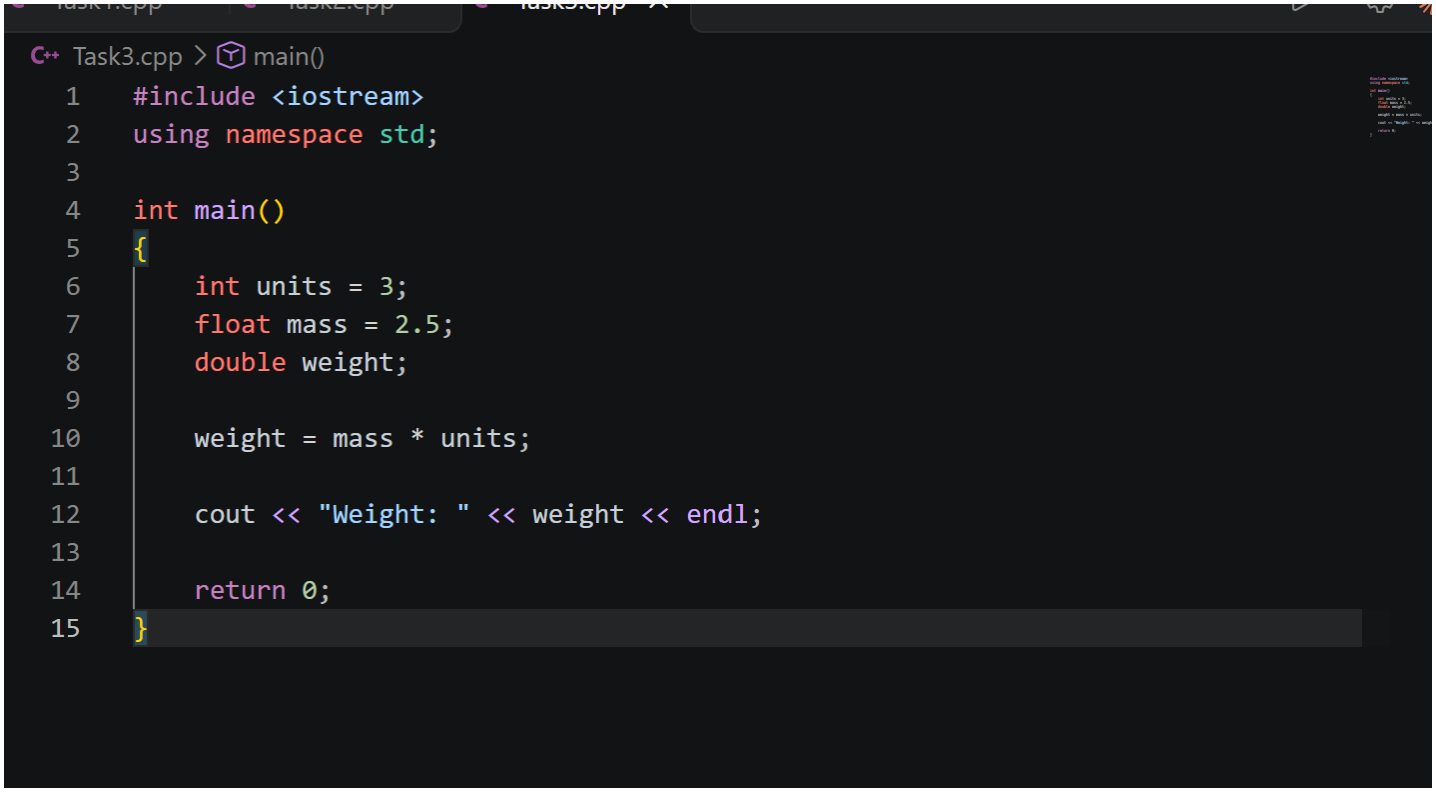
Which automatic data type conversion will take place?

A. mass is demoted to an int, units remains an int, and the result of mass * units is an int.

B. units is promoted to a float, mass remains a float, and the result of mass * units is a float.

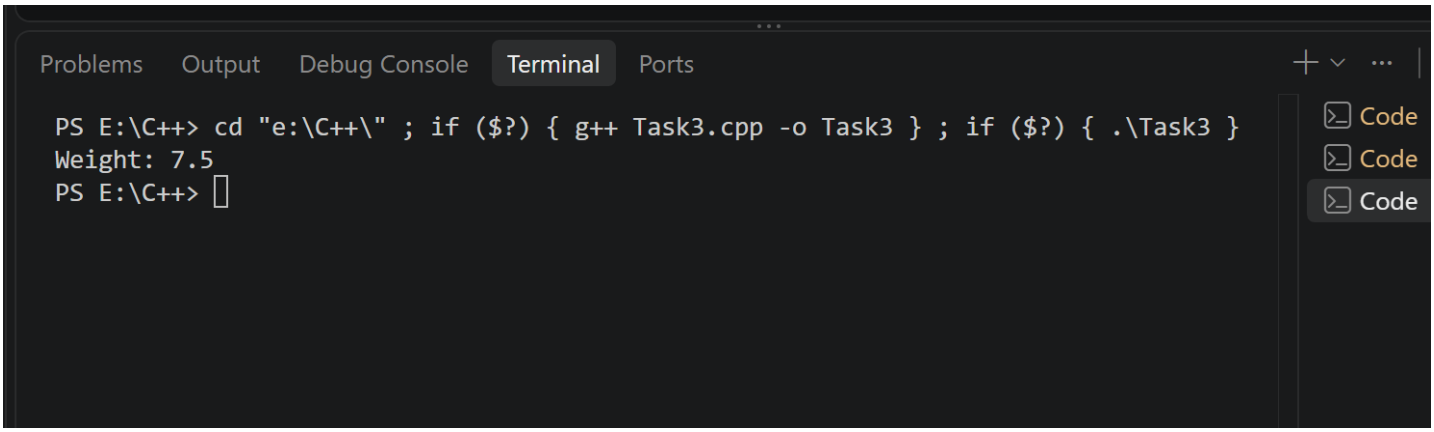
C. units is promoted to a float, mass remains a float, and the result of mass * units is a double.

Input



Handwritten Notes

Output


Problems
Output
Debug Console
Terminal
Ports
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ Task3.cpp -o Task3 } ; if ($?) { .\Task3 }
Weight: 7.5
PS E:\C++>
Code
Code
Code

Task 4:

Assume a program has the following variable definitions:

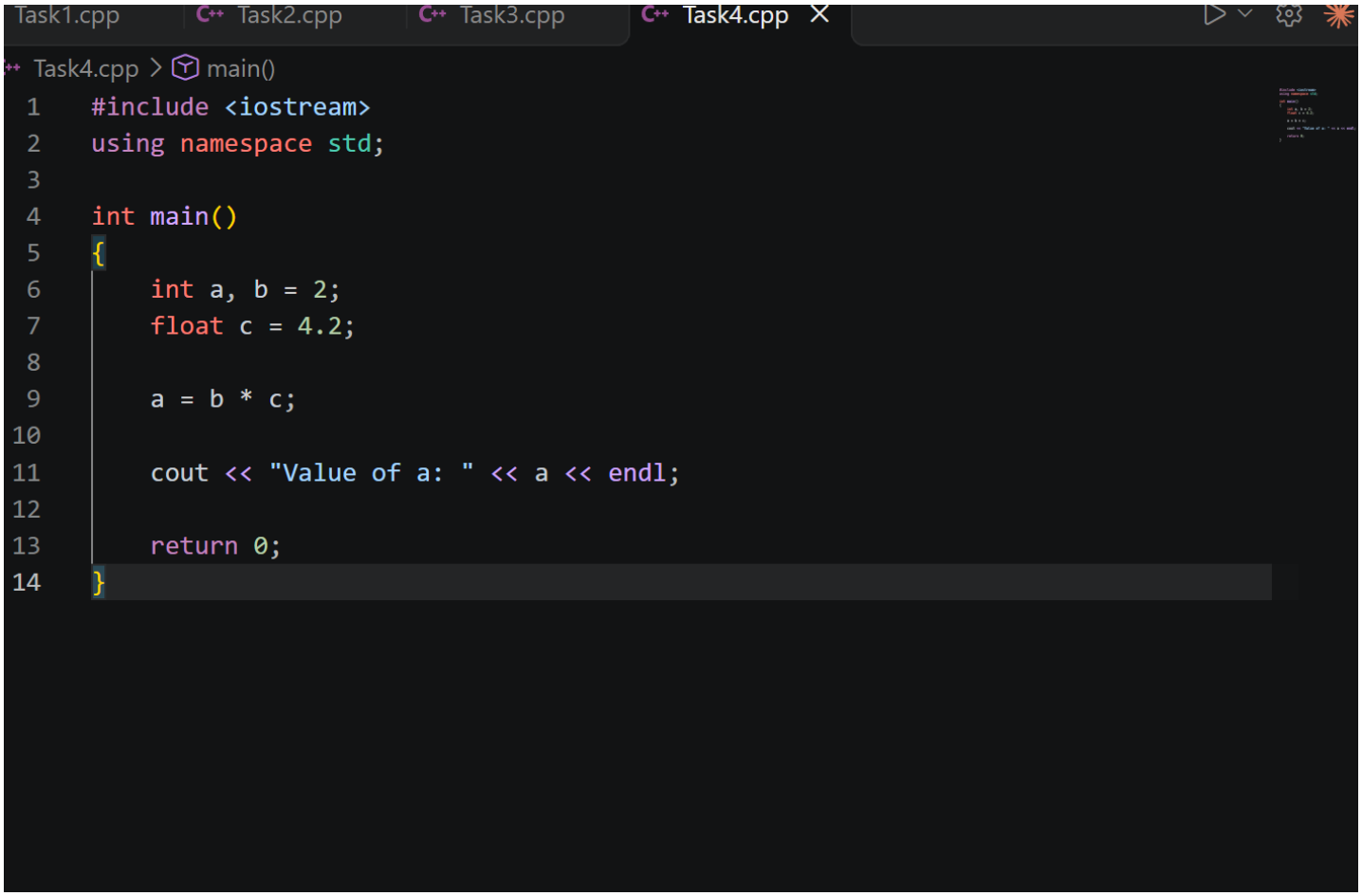
```
int a, b = 2;    float c = 4.2;
```

and the following statement: a = b * c;

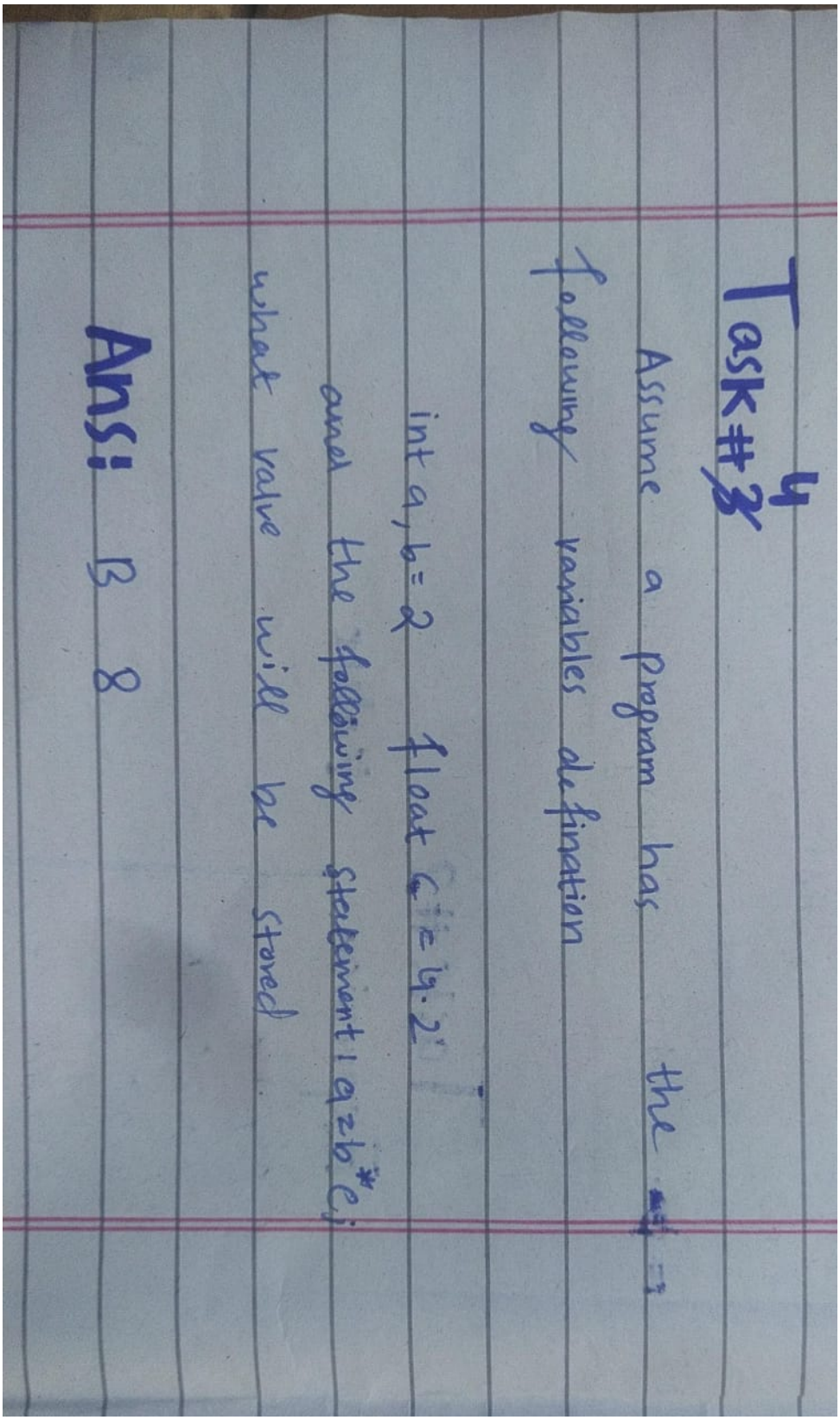
What value will be stored in a?

A. 8.4B. 8C. 0D. None of the above

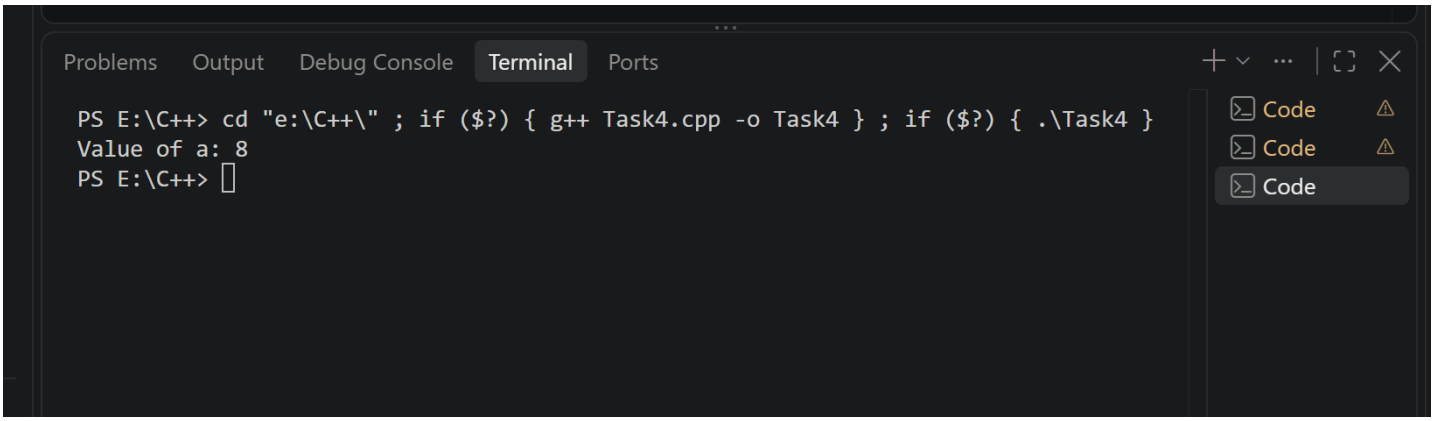
Input



Handwritten Notes



Output

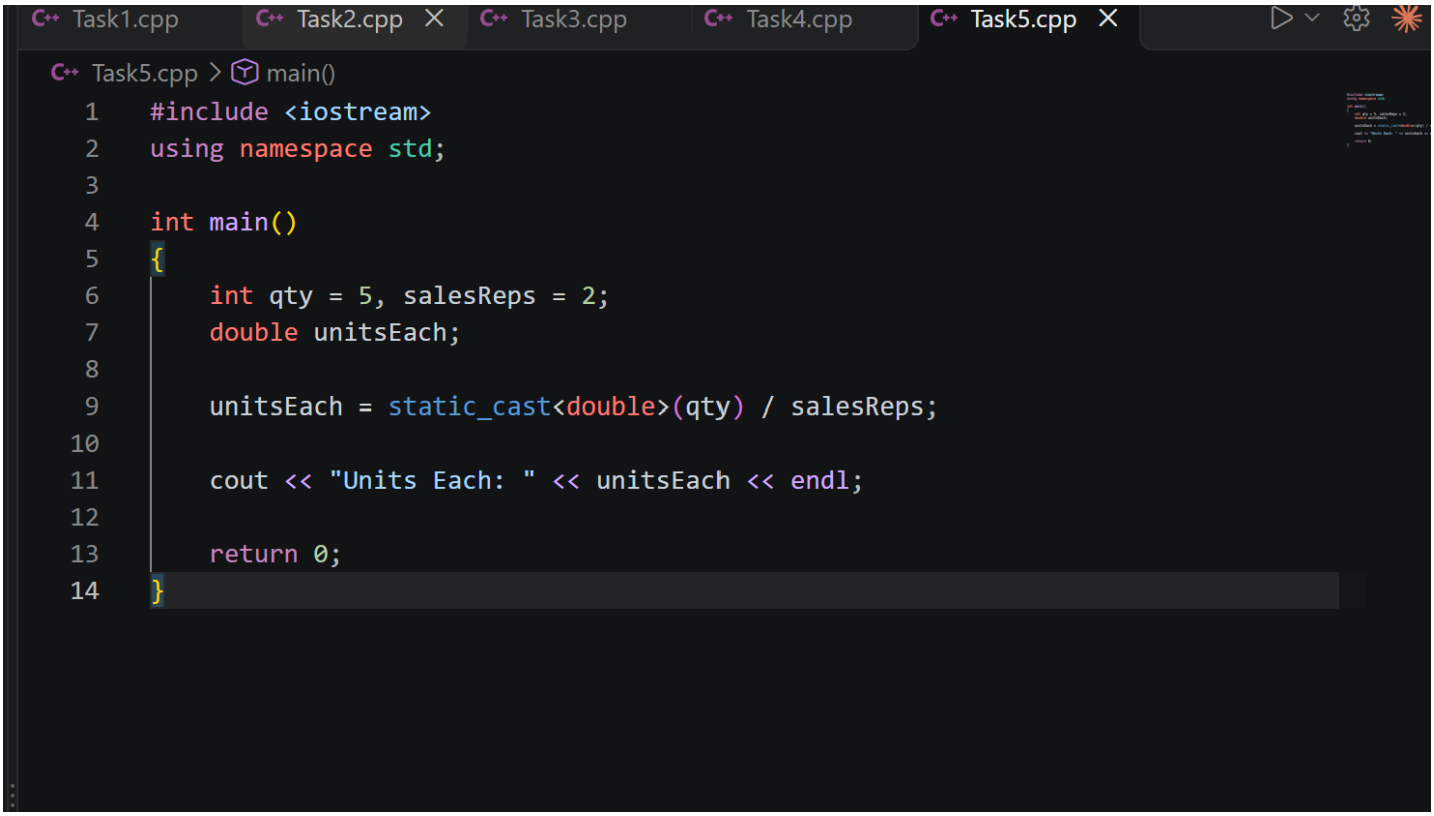

Problems
Output
Debug Console
Terminal
Ports
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ Task4.cpp -o Task4 } ; if ($?) { .\Task4 }
Value of a: 8
PS E:\C++>
Code
Code
Code

Task 5:

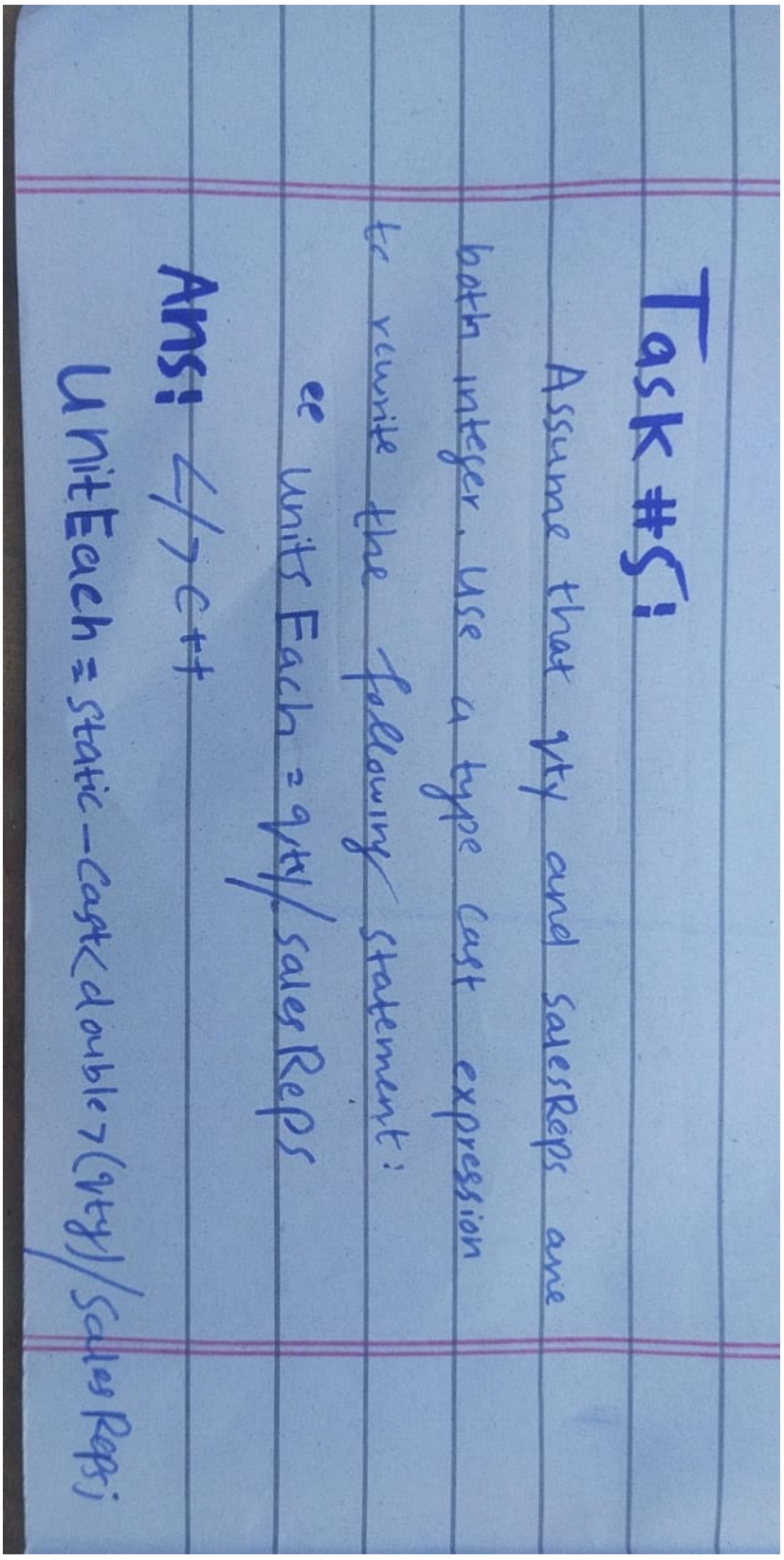
Assume that qty and salesReps are both integers. Use a type cast expression to rewrite the following statement so it will no longer perform integer division.

```
unitsEach = qty / salesReps;
```

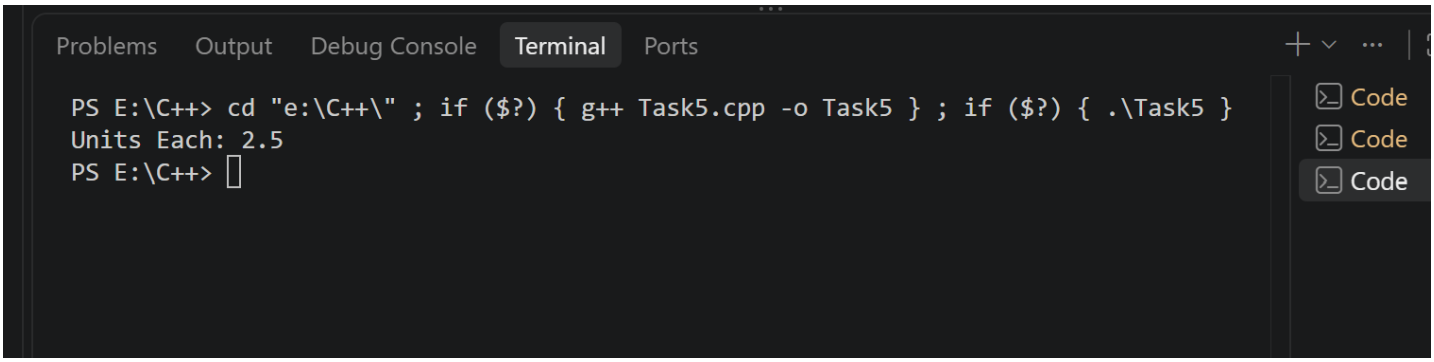
Input



Handwritten Notes



Output


Problems
Output
Debug Console
Terminal
Ports
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ Task5.cpp -o Task5 } ; if ($?) { .\Task5 }
Units Each: 2.5
PS E:\C++>
Code
Code
Code

Task 6:

Math Tutor: (hint rand())

Write a program that can be used as a math tutor for a young student. The program should display two random numbers to be added, such as

```
247
+ 129
---------
```

The program should then pause while the student works on the problem. When the student is ready to check the answer, he or she can press a key and the program will display the correct solution:

```
247
+ 129
---------
376
```

Input

Task6.cpp > main()

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main()
{
    srand(time(0));

    int num1 = rand() % 900 + 100;
    int num2 = rand() % 900 + 100;

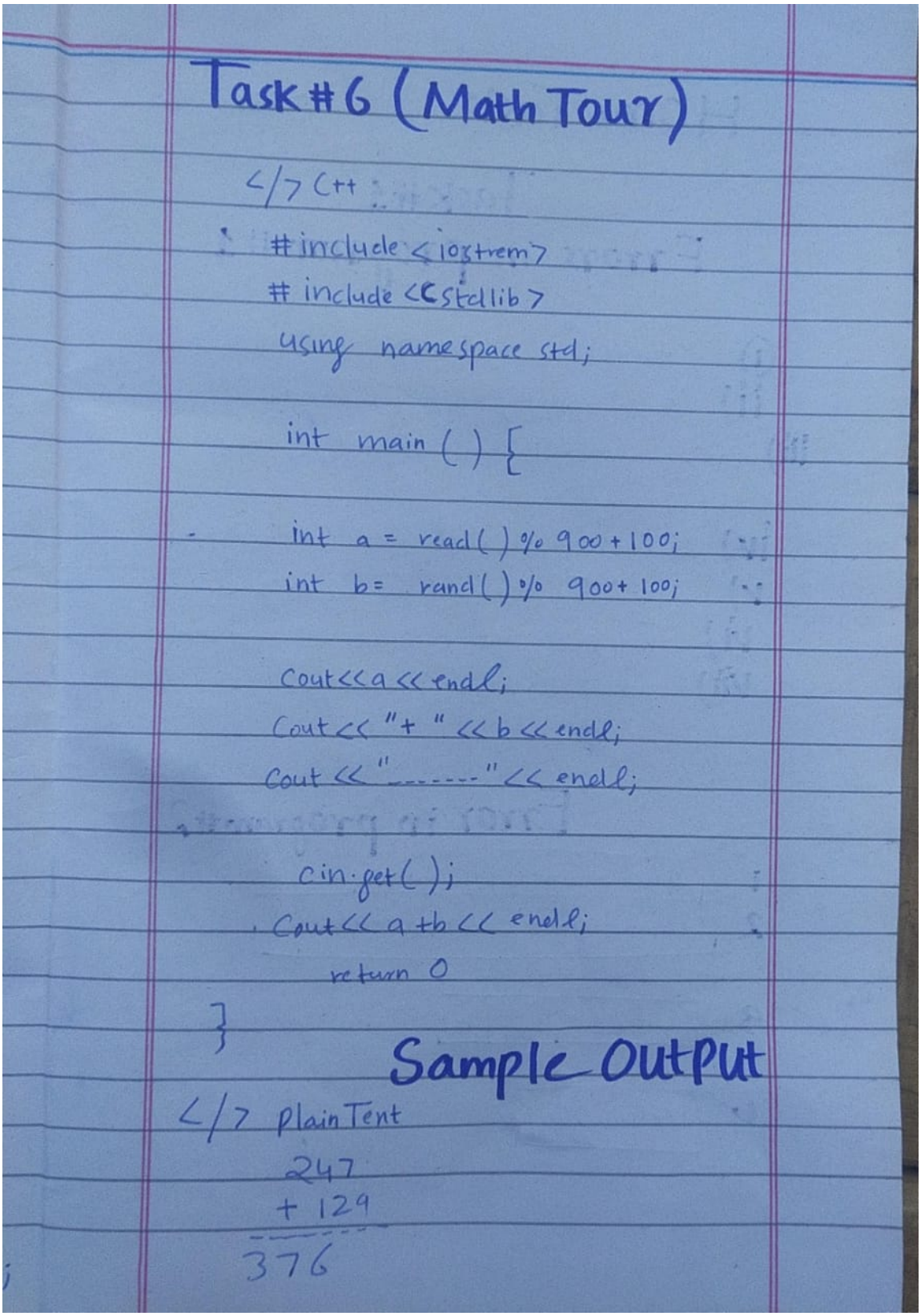
    cout << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "--------" << endl;

    cout << "Press Enter to see the answer...";
    cin.get();

    cout << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "--------" << endl;
    cout << num1 + num2 << endl;

    return 0;
}
```

Handwritten Notes



Task #6 (Math Tour)

</> C++

```
#include <iostream>
#include <cstdlib>
using namespace std;

int main ( ) {

    int a = rand( ) % 900 + 100;
    int b = rand( ) % 900 + 100;

    Cout<<a <<endl;
    Cout<< "+ " <<b <<endl;
    Cout << "------" << endl;

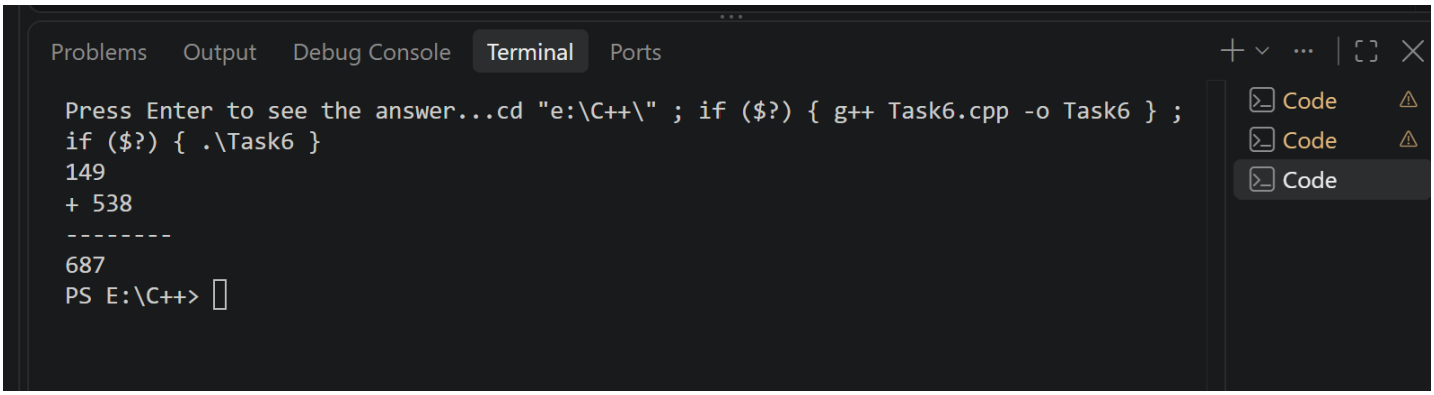
    cin.get( );
    Cout<< a+b << endl;
    return 0
}
```

Sample Output

</> PlainText

```
  247
+ 129
-----
  376
```

Output


Problems
Output
Debug Console
Terminal
Ports
Press Enter to see the answer...cd "e:\C++\" ; if ($?) { g++ Task6.cpp -o Task6 } ;
if ($?) { .\Task6 }
149
+ 538

687
PS E:\C++>
Code
Code
Code

Lab 4 (Home Tasks)

Task 1: (Program-1)

Each of the following programs has some errors. Locate as many as you can.

Program-1

```
using namespace std;
void main ()
{
■double number1, number2, sum;
■cout << "Enter a number: ";
■Cin << number1;
■cout << "Enter another number: ";
■cin << number2;
■number1 + number2 = sum;
■cout "The sum of the two numbers is " << sum
}
```

Input

Home_Task1_Program1.cpp > main()

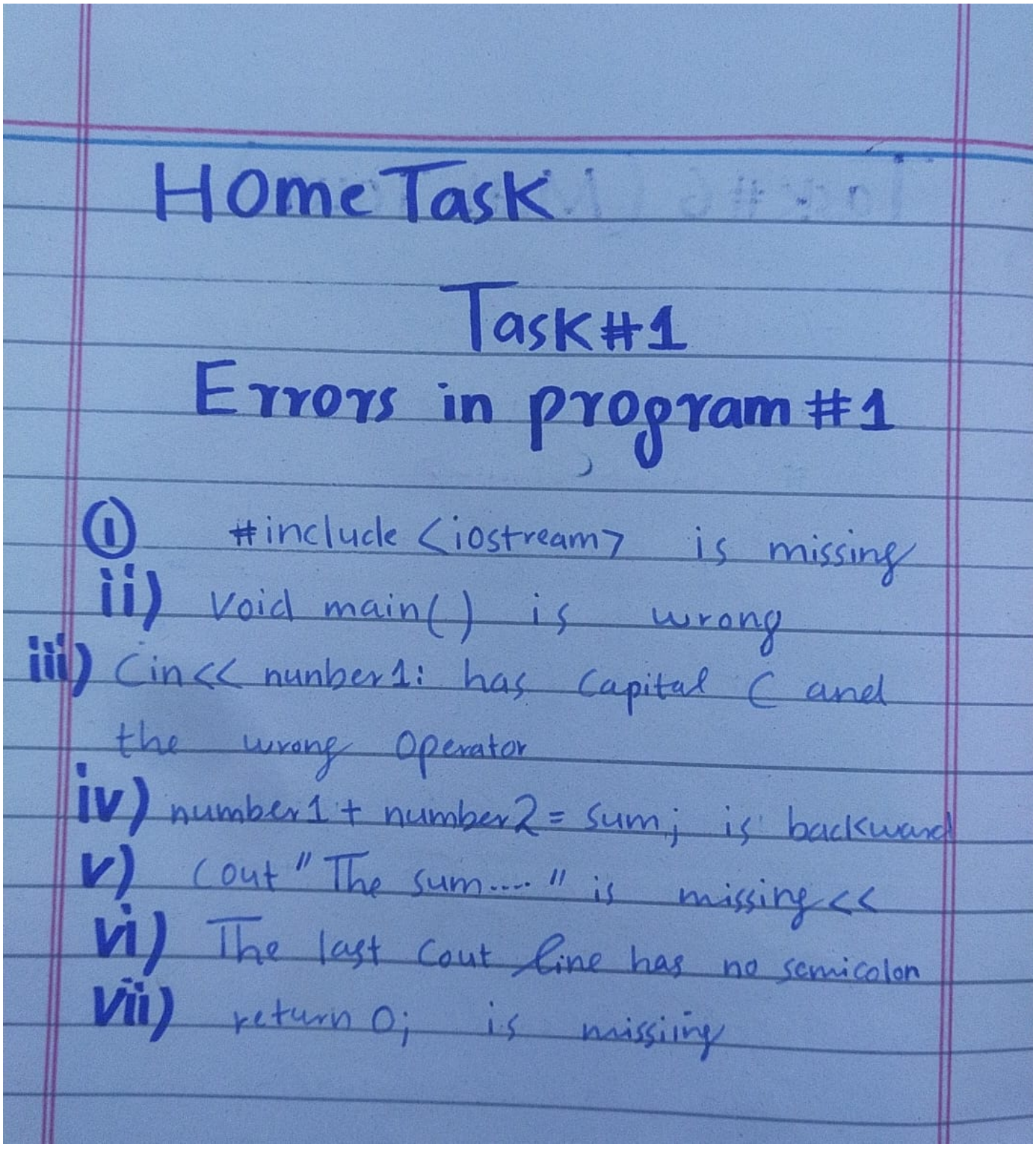
```cpp
#include <iostream>
using namespace std;

int main()
{
    double number1, number2, sum;

    cout << "Enter a number: ";
    cin >> number1;
    cout << "Enter another number: ";
    cin >> number2;

    sum = number1 + number2;
    cout << "The sum of the two numbers is " << sum;

    return 0;
}
```



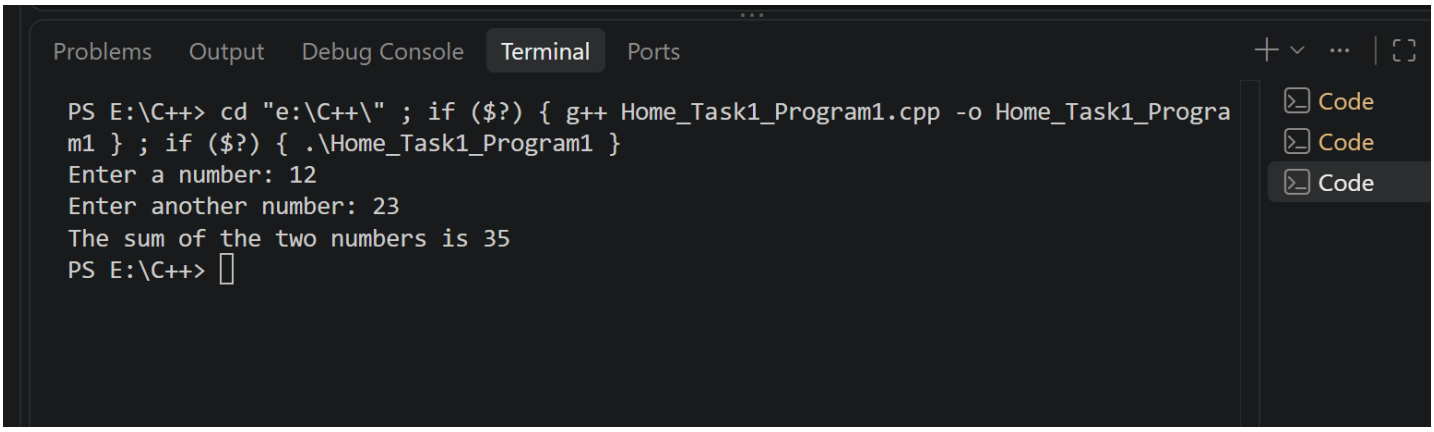
Home Task

Task#1

Errors in program#1

i) #include <iostream> is missing
ii) void main() is wrong
iii) Cin<< number1; has capital C and the wrong operator
iv) number1 + number2 = sum; is backward
v) Cout "The sum...." is missing <<
vi) The last Cout line has no semicolon
vii) return 0; is missing

Output


Problems
Output
Debug Console
Terminal
Ports
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ Home_Task1_Program1.cpp -o Home_Task1_Progra
m1 } ; if ($?) { .\Home_Task1_Program1 }
Enter a number: 12
Enter another number: 23
The sum of the two numbers is 35
PS E:\C++>
Code
Code
Code

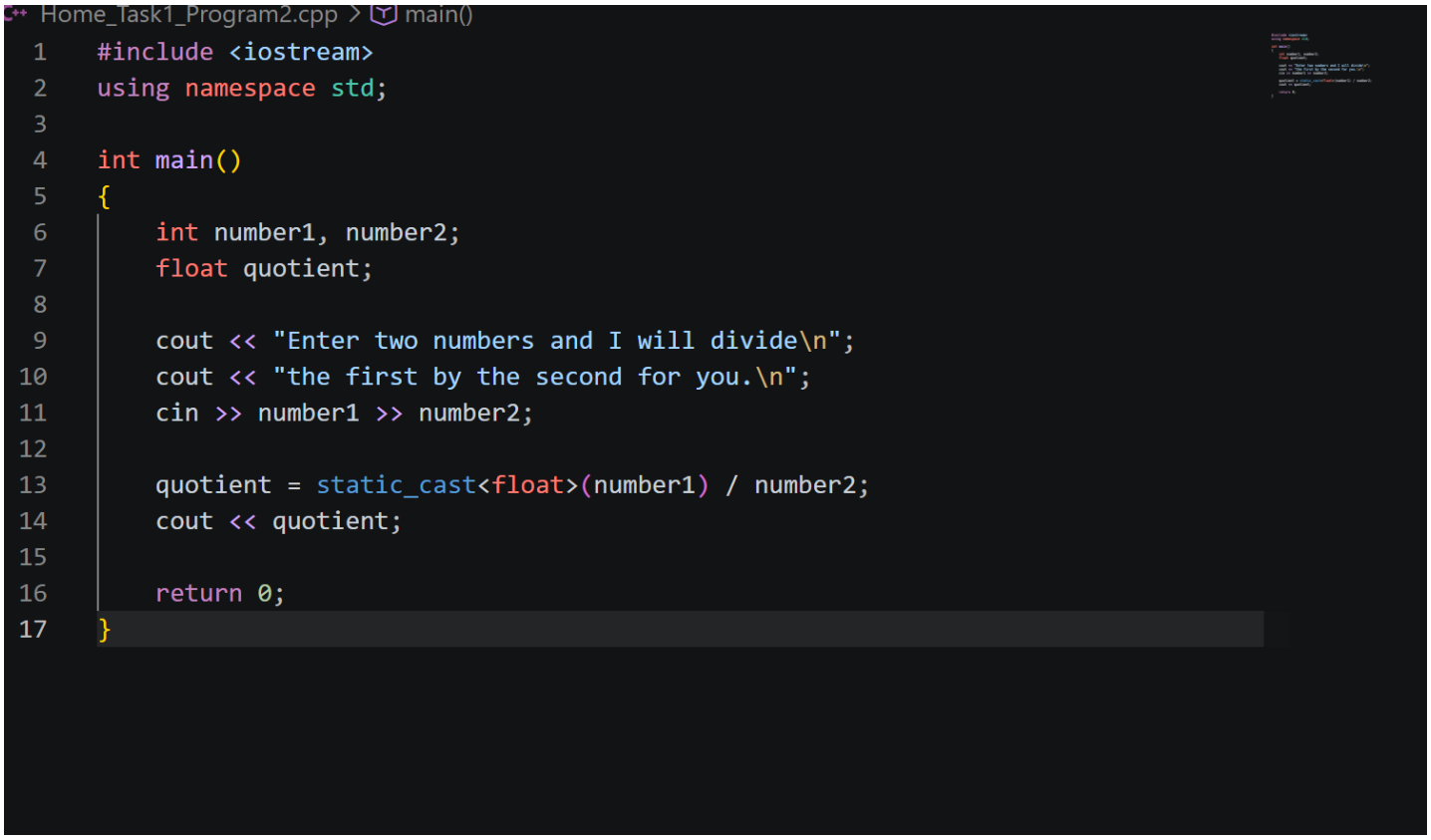
Task 1: (Program-2)

Each of the following programs has some errors. Locate as many as you can.

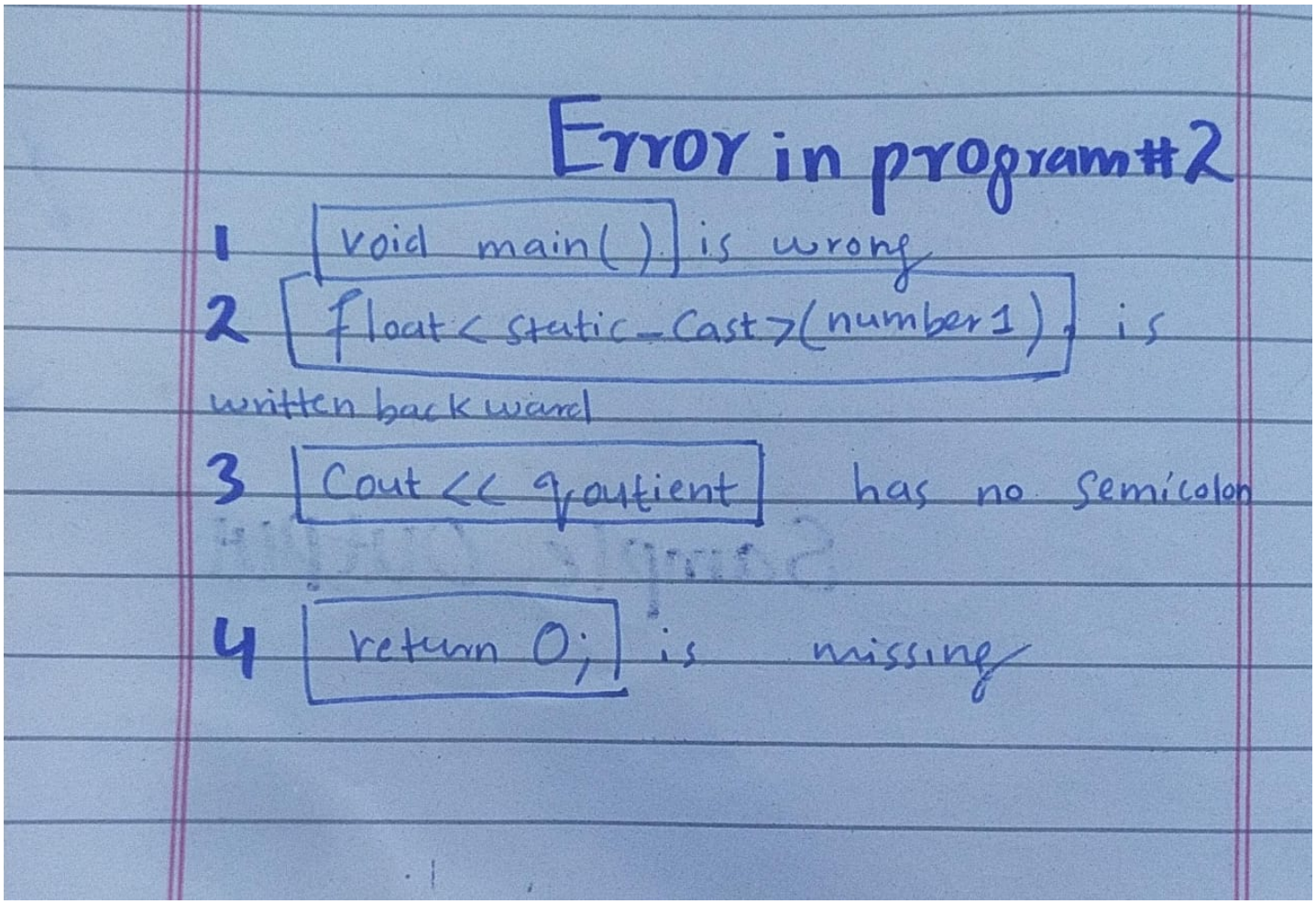
Program-2

```
#include <iostream>
using namespace std;
void main()
{
■int number1, number2;
■float quotient;
■cout << "Enter two numbers and I will divide\n";
■cout << "the first by the second for you.\n";
■cin >> number1, number2;
■quotient = float<static_cast>(number1) / number2;
■cout << quotient
}
```

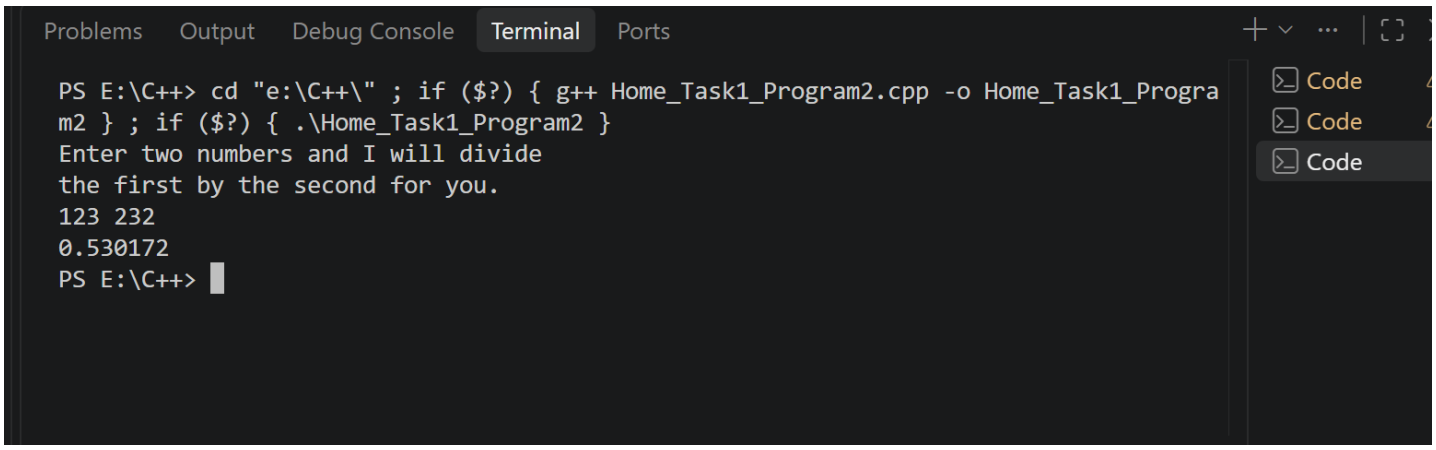
Input



Handwritten Notes



Output



Task 2:

Average of Values to get the average of a series of values, you add the values up and then divide the sum by the number of values. Write a program that stores the following values in five different variables: 28, 32, 37, 24, and 33. The program should first calculate the sum of these five variables and store the result in a separate variable named sum. Then, the program should divide the sum variable by 5 to get the average. Display the average on the screen.

Input

Home_Task2.cpp > main()

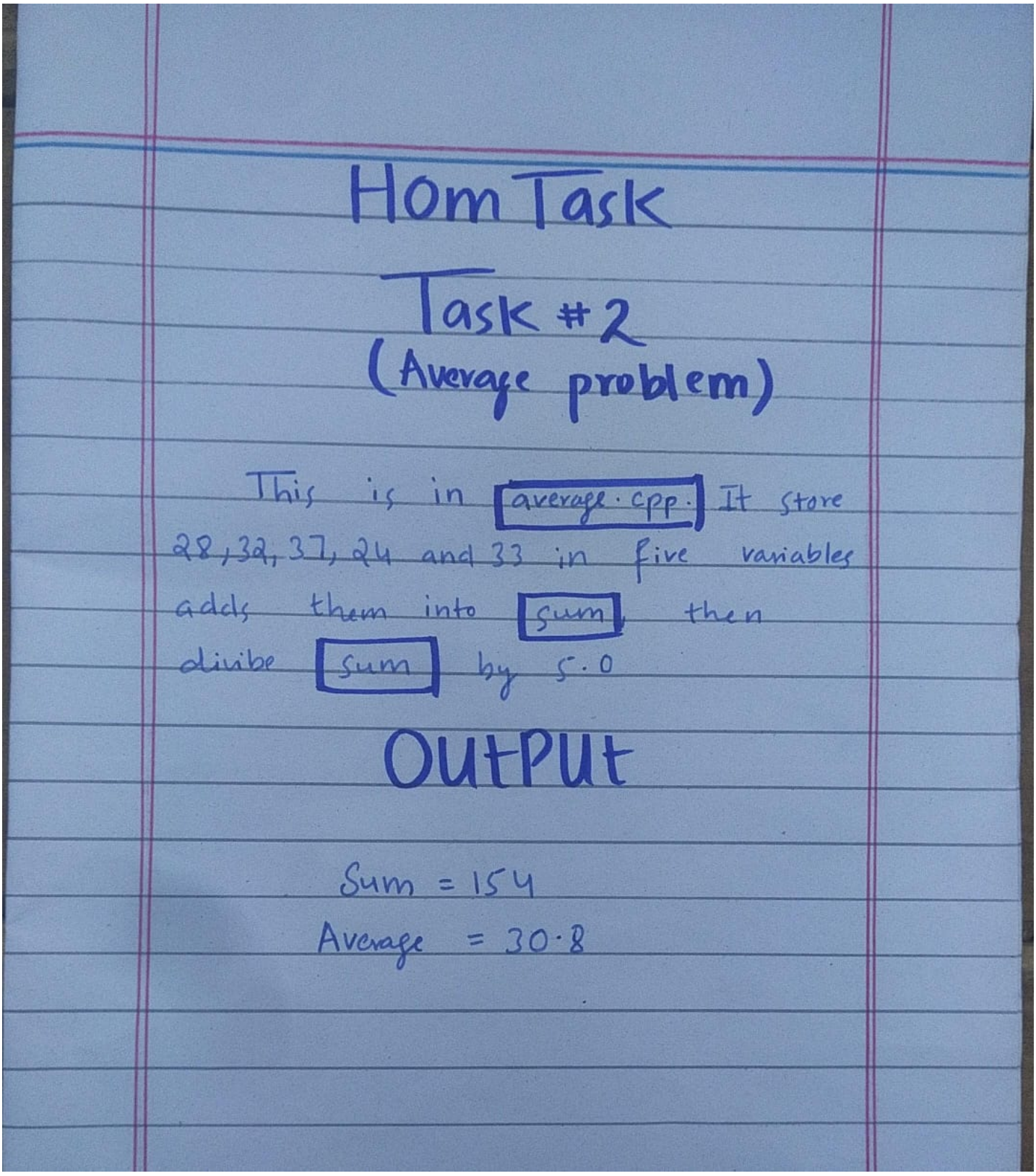
```cpp
#include <iostream>
using namespace std;

int main()
{
    int num1 = 28, num2 = 32, num3 = 37, num4 = 24, num5 = 33;

    int sum = num1 + num2 + num3 + num4 + num5;
    double average = sum / 5.0;

    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}
```



Hom Task

Task #2
(Average problem)

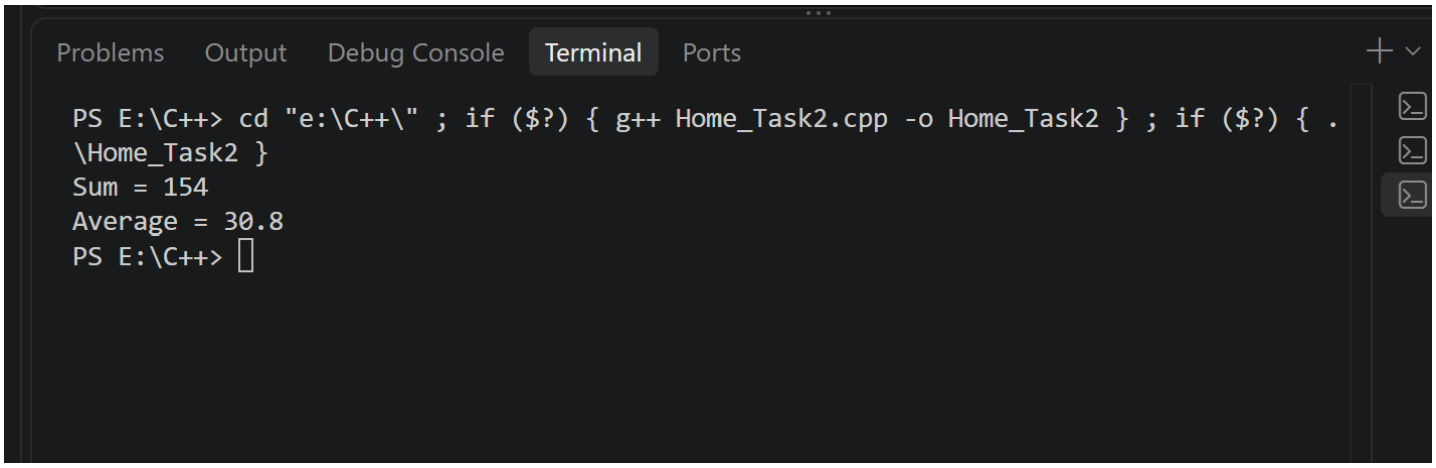
This is in `average.cpp`. It store 28, 32, 37, 24 and 33 in five variables adds them into `sum`, then divibe `sum` by 5.0

OutPut

Sum = 154

Average = 30.8

Output


Problems
Output
Debug Console
Terminal
Ports
PS E:\C++> cd "e:\C++\" ; if ($?) { g++ Home_Task2.cpp -o Home_Task2 } ; if ($?) { .
\Home_Task2 }
Sum = 154
Average = 30.8
PS E:\C++>